

P8131 Homework 4

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Housing Satisfaction in Copenhagen

```
htypes <- c("Tower", "Apartment", "House")
sat_lvl <- c("Low", "Medium", "High")

housing <- data.frame(
  housing_type = factor(rep(htypes, each = 2), levels = htypes),
  contact = factor(rep(c("Low", "High"), 3), levels = c("Low", "High")),
  low_sat = c(65, 34, 130, 141, 67, 130),
  med_sat = c(54, 47, 76, 116, 48, 105),
  high_sat = c(100, 100, 111, 191, 62, 104)
)
housing$total <- housing$low_sat + housing$med_sat + housing$high_sat
```

Problem 1

Summarize the data using appropriate tables of percentages to show the pair-wise associations between the levels of satisfaction and 1) contact with other residents and 2) type of housing. Comment on patterns.

Satisfaction by Contact Level

```
contact_tbl <- housing %>%
  group_by(contact) %>%
  summarize(Low = sum(low_sat), Medium = sum(med_sat),
            High = sum(high_sat), Total = sum(total)) %>%
  mutate(across(Low:High, ~ sprintf("%.1f%%", 100 * .x / Total),
            .names = "{.col}_pct"))

kable(contact_tbl %>% dplyr::select(contact, Low, Low_pct, Medium, Medium_pct,
                                High, High_pct, Total),
      col.names = c("Contact", "Low", "(%)", "Medium", "(%)",
                    "High", "(%)", "Total"),
      caption = "Satisfaction by Contact Level")
```

Table 1: Satisfaction by Contact Level

Contact	Low	(%)	Medium	(%)	High	(%)	Total
Low	262	36.7%	178	25.0%	273	38.3%	713
High	305	31.5%	268	27.7%	395	40.8%	968

Pattern: High-contact residents report higher satisfaction (40.8% high) compared to low-contact residents (38.3% high). Low satisfaction is more prevalent among low-contact residents (36.7%) than high-contact (31.5%). More contact with neighbors is associated with greater housing satisfaction.

Satisfaction by Housing Type

```
housing_tbl <- housing %>%
  group_by(housing_type) %>%
  summarize(Low = sum(low_sat), Medium = sum(med_sat),
            High = sum(high_sat), Total = sum(total)) %>%
  mutate(across(Low:High, ~ sprintf("%.1f%%", 100 * .x / Total),
            .names = "{.col}_pct"))

kable(housing_tbl %>% dplyr::select(housing_type, Low, Low_pct, Medium,
                                Medium_pct, High, High_pct, Total),
      col.names = c("Housing Type", "Low", "(%)", "Medium", "(%)",
                    "High", "(%)", "Total"),
      caption = "Satisfaction by Housing Type")
```

Table 2: Satisfaction by Housing Type

Housing Type	Low	(%)	Medium	(%)	High	(%)	Total
Tower	99	24.8%	101	25.2%	200	50.0%	400
Apartment	271	35.4%	192	25.1%	302	39.5%	765
House	197	38.2%	153	29.7%	166	32.2%	516

Pattern: Tower block residents have the highest proportion of high satisfaction (50.0%) and lowest low satisfaction (24.8%). House residents show the most low satisfaction (38.2%) and least high satisfaction (32.2%). Apartment residents fall in between.

Problem 2

Use nominal logistic regression to model the associations. Obtain a model that summarizes the patterns. Describe findings (odds ratios with 95% CIs, goodness-of-fit). Is there interaction?

We use satisfaction (Low/Medium/High) as the response with “Low” as reference:

$$\log \frac{\pi_j}{\pi_{\text{Low}}} = \beta_{0j} + \beta_{1j} \text{Contact}_{\text{High}} + \beta_{2j} \text{Apartment} + \beta_{3j} \text{House}, \quad j \in \{\text{Med, High}\}$$

```
# Reshape to long format
dat <- data.frame(
  housing_type = rep(housing$housing_type, 3),
  contact = rep(housing$contact, 3),
  satisfaction = factor(rep(sat_lvl, each = 6), levels = sat_lvl),
  count = c(housing$low_sat, housing$med_sat, housing$high_sat)
)
```

Test for Interaction

```
fit_full <- multinom(satisfaction ~ contact * housing_type,
                     weights = count, data = dat, trace = FALSE)
fit_main <- multinom(satisfaction ~ contact + housing_type,
                     weights = count, data = dat, trace = FALSE)
summary(fit_full)
```

```
## Call:
## multinom(formula = satisfaction ~ contact * housing_type, data = dat,
##           weights = count, trace = FALSE)
##
## Coefficients:
##           (Intercept) contactHigh housing_typeApartment housing_typeHouse
## Medium -0.1854029    0.5091910          -0.3513970          -0.1480892
## High    0.4307831    0.6480263          -0.5887874          -0.5083426
##           contactHigh:housing_typeApartment contactHigh:housing_typeHouse
## Medium                                -0.1675605                        -0.3892720
## High                                -0.1865089                        -0.7936097
##
## Std. Errors:
##           (Intercept) contactHigh housing_typeApartment housing_typeHouse
## Medium  0.1841280    0.2908461          0.2339943          0.2639353
## High    0.1593255    0.2545513          0.2051486          0.2375690
##           contactHigh:housing_typeApartment contactHigh:housing_typeHouse
## Medium                                0.3480726                        0.3708993
## High                                0.3063095                        0.3363904
##
## Residual Deviance: 3598.587
## AIC: 3622.587
```

```
summary(fit_main)
```

```
## Call:
## multinom(formula = satisfaction ~ contact + housing_type, data = dat,
##           weights = count, trace = FALSE)
##
## Coefficients:
##           (Intercept) contactHigh housing_typeApartment housing_typeHouse
## Medium -0.1072644    0.2959803          -0.4067537          -0.3370771
## High    0.5607737    0.3282263          -0.6415967          -0.9456177
##
```

```
## Std. Errors:
##      (Intercept) contactHigh housing_typeApartment housing_typeHouse
## Medium    0.1524077    0.1301046                0.1713011        0.1803577
## High      0.1329301    0.1181870                0.1500773        0.1644850
##
## Residual Deviance: 3605.48
## AIC: 3621.48
```

```
dev_diff <- fit_main$deviance - fit_full$deviance
df_diff <- length(coef(fit_full)) - length(coef(fit_main))
p_int <- 1 - pchisq(dev_diff, df_diff)
```

```
cat("LRT for interaction: G =", round(dev_diff, 3),
    ", df =", df_diff, ", p =", round(p_int, 3), "\n")
```

```
## LRT for interaction: G = 6.893 , df = 4 , p = 0.142
```

The interaction is not significant ($p = 0.142$), so we use the main effects model.

```
# Odds ratios with 95% CIs
coefs <- coef(fit_main)
se <- summary(fit_main)$standard.errors
OR <- exp(coefs)
OR_lo <- exp(coefs - 1.96 * se)
OR_hi <- exp(coefs + 1.96 * se)

for (j in c("Medium", "High")) {
  cat("\n--- ", j, "vs Low ---\n")
  res <- data.frame(OR = round(OR[j, ], 3),
                    CI_lower = round(OR_lo[j, ], 3),
                    CI_upper = round(OR_hi[j, ], 3))
  print(kable(res))
}
```

```
##
## --- Medium vs Low ---
##
##
## |                |      OR| CI_lower| CI_upper|
## |:-----:|-----:|-----:|-----:|
## |(Intercept)    | 0.898|    0.666|    1.211|
## |contactHigh    | 1.344|    1.042|    1.735|
## |housing_typeApartment | 0.666|    0.476|    0.931|
## |housing_typeHouse | 0.714|    0.501|    1.017|
##
## --- High vs Low ---
##
##
## |                |      OR| CI_lower| CI_upper|
## |:-----:|-----:|-----:|-----:|
## |(Intercept)    | 1.752|    1.350|    2.273|
## |contactHigh    | 1.389|    1.101|    1.750|
## |housing_typeApartment | 0.526|    0.392|    0.706|
## |housing_typeHouse | 0.388|    0.281|    0.536|
```

Goodness-of-Fit

```
patterns <- distinct(dat[, c("housing_type", "contact")])
obs <- matrix(dat$count, nrow = 6, ncol = 3)
m <- rowSums(obs)
pred <- predict(fit_main, newdata = patterns, type = "probs")
expected <- pred * m

df_gof <- 6 * 2 - 8 # 6 patterns x (3-1) - 8 parameters = 4

# Pearson chi-square
X2 <- sum((obs - expected)^2 / expected)
p_X2 <- 1 - pchisq(X2, df_gof)

# Deviance (likelihood ratio)
D <- 2 * sum(obs * log(obs / expected))
p_D <- 1 - pchisq(D, df_gof)

cat("Pearson X2 =", round(X2, 3), ", df =", df_gof, ", p =", round(p_X2, 3), "\n")

## Pearson X2 = 6.932 , df = 4 , p = 0.14

cat("Deviance   =", round(D, 3), ", df =", df_gof, ", p =", round(p_D, 3), "\n")

## Deviance   = 6.893 , df = 4 , p = 0.142
```

Interpretation

1. **Contact:** High contact significantly increases odds of both medium (OR = 1.34, 95% CI: 1.04–1.73) and high satisfaction (OR = 1.39, 95% CI: 1.1–1.75) vs. low satisfaction.
2. **Housing type:** Compared to tower blocks, apartment residents have lower odds of high satisfaction (OR = 0.53, 95% CI: 0.39–0.71), and house residents have even lower odds (OR = 0.39, 95% CI: 0.28–0.54).
3. **Goodness-of-fit:** Pearson $\chi^2 = 6.93$ ($p = 0.14$) and Deviance = 6.89 ($p = 0.142$) with 4 df, both indicating adequate fit.
4. **No significant interaction** ($p = 0.142$).

Problem 3

Fit a proportional odds model with the same variables. What does it tell?

The proportional odds model:

$$\log \frac{P(Y \leq j)}{P(Y > j)} = \beta_{0j} - \mathbf{x}^T \beta, \quad j = 1, 2$$

```

dat$satisfaction_ord <- ordered(dat$satisfaction, levels = sat_lvl)
fit_po <- polr(satisfaction_ord ~ contact + housing_type,
              weights = count, data = dat)
summary(fit_po)

## Call:
## polr(formula = satisfaction_ord ~ contact + housing_type, data = dat,
##       weights = count)
##
## Coefficients:
##               Value Std. Error t value
## contactHigh      0.2524   0.09306   2.713
## housing_typeApartment -0.5009   0.11675  -4.291
## housing_typeHouse   -0.7362   0.12610  -5.838
##
## Intercepts:
##               Value Std. Error t value
## Low|Medium  -0.9973   0.1075   -9.2794
## Medium|High   0.1152   0.1047    1.1004
##
## Residual Deviance: 3610.286
## AIC: 3620.286

# Cumulative ORs for higher satisfaction
# polr: logit(P(Y<=j)) = beta_0j - x'beta, so exp(beta) = OR for Y > j
b <- coef(fit_po)
se_po <- sqrt(diag(vcov(fit_po)))[1:length(b)]

OR_po <- exp(b)
CI_lo_po <- exp(b - 1.96 * se_po)
CI_hi_po <- exp(b + 1.96 * se_po)

po_res <- data.frame(
  Coefficient = round(b, 4), SE = round(se_po, 4),
  OR = round(OR_po, 3),
  CI_lower = round(CI_lo_po, 3),
  CI_upper = round(CI_hi_po, 3)
)
kable(po_res, col.names = c("Coef", "SE", "OR (higher sat.)",
                           "95% CI Lower", "95% CI Upper"),
      caption = "Proportional Odds Model: Cumulative ORs")

```

Table 3: Proportional Odds Model: Cumulative ORs

	Coef	SE	OR (higher sat.)	95% CI Lower	95% CI Upper
contactHigh	0.2524	0.0931	1.287	1.073	1.545
housing_typeApartment	-0.5009	0.1168	0.606	0.482	0.762
housing_typeHouse	-0.7362	0.1261	0.479	0.374	0.613

```

# Goodness-of-fit
pred_po <- predict(fit_po, newdata = patterns, type = "probs")
exp_po <- pred_po * m
df_po <- 6 * 2 - 5 # 12 - 5 parameters = 7

X2_po <- sum((obs - exp_po)^2 / exp_po)
p_X2_po <- 1 - pchisq(X2_po, df_po)

D_po <- 2 * sum(obs * log(obs / exp_po))
p_D_po <- 1 - pchisq(D_po, df_po)

cat("Pearson  $\chi^2$  =", round(X2_po, 3), ", df =", df_po, ", p =", round(p_X2_po, 3), "\n")

## Pearson  $\chi^2$  = 11.642 , df = 7 , p = 0.113

cat("Deviance   =", round(D_po, 3), ", df =", df_po, ", p =", round(p_D_po, 3), "\n")

## Deviance   = 11.699 , df = 7 , p = 0.111

```

Interpretation

1. **Contact:** High contact increases odds of higher satisfaction by a factor of 1.29 (95% CI: 1.07–1.54).
2. **Housing type:** Apartment residents have lower odds of higher satisfaction vs. tower blocks (OR = 0.61, 95% CI: 0.48–0.76). House residents have even lower odds (OR = 0.48, 95% CI: 0.37–0.61).
3. **Model fit:** Pearson χ^2 = 11.64 (p = 0.113) and Deviance = 11.7 (p = 0.111) with 7 df, both indicating adequate fit.
4. **Parsimony:** The proportional odds model uses 5 parameters vs. 8 in the nominal model, giving a more concise summary while fitting the data well.

Problem 4

Calculate Pearson residuals from the proportional odds model. Where are the largest discrepancies?

```

resid_po <- (obs - exp_po) / sqrt(exp_po)

rownames(resid_po) <- paste(rep(htypes, each = 2),
                           rep(c("Low", "High"), 3), sep = "-")
colnames(resid_po) <- c("Low Sat", "Med Sat", "High Sat")

kable(round(resid_po, 3), caption = "Pearson Residuals")

```

Table 4: Pearson Residuals

	Low Sat	Med Sat	High Sat
Tower-Low	0.779	-0.370	-0.315
Tower-High	-0.995	0.455	0.335
Apartment-Low	0.918	-1.067	-0.015
Apartment-High	-0.237	-0.405	0.538
House-Low	-1.141	0.140	1.244
House-High	0.274	1.368	-1.478

```
cat("Range of residuals:", round(range(resid_po), 3), "\n")
```

```
## Range of residuals: -1.478 1.368
```

```
cat("Largest |residual|:", round(max(abs(resid_po)), 3), "\n")
```

```
## Largest |residual|: 1.478
```

Interpretation

All Pearson residuals are relatively small ($|r| < 1.5$), confirming adequate model fit. The largest discrepancies involve house residents: the model over-predicts high satisfaction for high-contact house residents ($r = -1.48$) and under-predicts medium satisfaction for the same group ($r = 1.37$). This suggests minor heterogeneity in the contact-by-housing-type pattern that the main effects model does not fully capture.